



CMPSC 445: Machine Learning Data Science

Project-2

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TABLE OF CONTENTS

1	Introduction / Project Description	3
2	Significance of the Project	3
3	Data Collection	3
4	Data Processing and Feature Engineering	4
5	Model Development	5
6	Showcase	5
7	Discussion and Conclusion	7
8	AI Usage	8

1. Introduction / Project Description

For this project, we built a web-based machine learning application called Best PA County to Live In Predictor. The main goal of the project is to compare Pennsylvania counties and predict a livability score for each county. Since the project theme is “Best in PA,” we decided to define “best” as the county that has stronger overall conditions in areas like income, poverty, health, education, housing, and community access.

The app lets a user choose a Pennsylvania county from a dropdown menu. After the county is selected, the app shows a predicted livability score, the county’s rank in Pennsylvania, a category such as Excellent or Good, and a summary of important county features. The app also includes a full ranking table so users can compare all counties.

2. Significance of the Project

This project is meaningful because people usually think about many different factors when deciding whether a place is good to live in. A county might have high income, but it could also have housing problems. Another county might have good health outcomes but fewer economic opportunities. Because of that, looking at only one factor is not enough.

Our project tries to combine several important factors into one livability score. This makes it easier for users to compare counties without having to read a large spreadsheet. The app gives a quick result, but it also shows supporting features so the user can understand why a county scored higher or lower.

The project is also interesting because it turns public data into something more interactive. Instead of only showing numbers in a table, we used data processing, feature engineering, machine learning, and a web app to make the information easier to use.

3. Data Collection

We used two public data sources for this project.

The first data source is [the 2025 Pennsylvania County Health Rankings Excel workbook](#). This file includes county-level information about health, education, housing, community access, unemployment, broadband access, and other quality-of-life measures. The workbook has several sheets, including Select Measure Data, Additional Measure Data, Select Measure Sources & Years, and Health Groups. We mainly used the Select Measure Data and Additional Measure Data sheets because they had the county-level features needed for the model.

The second data source is [the PA Open Data ACS Income and Poverty county dataset](#). This dataset includes income and poverty information from the U.S. Census ACS estimates. It has 469 rows and 27 columns. Some useful columns include county name, population, population below poverty, median household income, and per capita income.

Sample Data

Here is a small sample from the ACS income and poverty dataset:

[illegible]

4. Data Processing and Feature Engineering

The first step in data processing was loading the Excel and CSV files into Python. For the County Health Rankings workbook, we used the Select Measure Data and Additional Measure Data sheets. We removed the statewide Pennsylvania row because we only wanted to compare counties. Then we converted numeric columns into number format so they could be used for calculations and model training.

For the ACS dataset, the year column was converted into a date format. Since this dataset contains multiple years, we selected the latest year available. We also created a full Pennsylvania county FIPS code by adding the county code to 42000. This allowed us to merge the ACS data with the County Health Rankings data using the FIPS column.

After merging both datasets, we selected features related to five main livability areas:

- **Economic factors:** median household income, per capita income, poverty rate, unemployment rate, and income inequality.
- **Health factors:** life expectancy, poor or fair health, physically unhealthy days, mentally unhealthy days, and uninsured rate.
- **Education factors:** some college and high school completion.
- **Community access factors:** access to exercise opportunities, food environment index, broadband access, and social association rate.
- **Housing and family factors:** severe housing problems, child care cost burden, and children in poverty.

For missing values, we filled numeric missing data with the median of that column. We used the median because it is less affected by extreme values than the average.

Next, we created a custom livability score. Each feature was converted into a 0–100 score using min-max scaling. For positive features, such as income, life expectancy, and education, higher values got higher scores. For negative features, such as poverty, unemployment, housing problems, and poor health, the scores were reversed so lower values got better scores.

The final livability score was calculated like this:

```
model_df["livability_score"] = (  
    0.30 * model_df["economic_score"]  
    + 0.25 * model_df["health_score"]  
    + 0.20 * model_df["education_score"]  
    + 0.15 * model_df["community_access_score"]  
    + 0.10 * model_df["housing_family_score"]  
)
```

We gave economic factors the highest weight because income, poverty, and employment are major parts of quality of life. Health and education were also weighted strongly because they are important long-term indicators.

5. Model Development

The machine learning task for this project is regression. The model takes county-level features as input and predicts a livability score as output.

The input features include median household income, per capita income, poverty rate, population, life expectancy, access to exercise, food environment index, education indicators, broadband access, unemployment rate, children in poverty, severe housing problems, uninsured rate, poor health indicators, income inequality, child care cost burden, and rural percentage.

We tested three regression models:

Model Comparison

Model	MAE	RMSE	R ² Score
Linear Regression	0	0	1
Random Forest Regressor	2.9894	3.6315	0.9143
Decision Tree Regressor	4.3888	5.1822	0.8256

Linear Regression performed the best based on the test results. Its R² score was 1.0000, and its MAE and RMSE were almost zero. This makes sense because our livability score was created from a weighted combination of numeric features. Since Linear Regression is good at learning linear relationships, it was able to learn the scoring pattern almost exactly. Random Forest also performed well, with an R² score of 0.9143. Decision Tree had the weakest result out of the three models. Based on the results, Linear Regression was selected as the final model for the web app.

6. Showcase

The final software is a Streamlit web app called Best PA County to Live In Predictor. The model is trained separately in Python, and the app loads the trained model for prediction. This means the app is used for inference, not for training.

The app has four main tabs:

County Prediction:

- The user selects a Pennsylvania county from the sidebar. The app then shows the predicted livability score, state rank, category, and a summary of important features for that county.

County Selection

Select a Pennsylvania County

Philadelphia

Project Theme: Best in PA

ML Task: Regression

Model: Machine Learning Model

Output: Livability Score

Best PA County to Live In Predictor

This web app uses a trained machine learning model to predict and rank Pennsylvania counties based on livability-related indicators. The score is based on economic, health, education, housing, and community access factors.

County Prediction

Rankings

Model Details

Project Notes

Prediction Result: Philadelphia County

Predicted Livability Score

36.28

State Rank

66 / 67

Category

Needs Improvement

County Feature Summary

Feature	Value
Median Household Income	\$45,927
Per Capita Income	\$27,924
Poverty Rate	24.34%
Life Expectancy	73.94
Unemployment Rate	4.23%
Children in Poverty	25.80%
Some College	68.80%
High School Completion	87.40%
Broadband Access	87.50%
Severe Housing Problems	22.26%

The score is not an official government ranking. It is a project-defined machine learning score based on selected public county-level data.

Rankings:

- This tab shows the top 10 counties and the full ranking table for all Pennsylvania counties. There is also a button to download the rankings as a CSV file.

County Selection

Select a Pennsylvania County

Philadelphia

Project Theme: Best in PA

ML Task: Regression

Model: Machine Learning Model

Output: Livability Score

Best PA County to Live In Predictor

This web app uses a trained machine learning model to predict and rank Pennsylvania counties based on livability-related indicators. The score is based on economic, health, education, housing, and community access factors.

County Prediction

Rankings

Model Details

Project Notes

Pennsylvania County Rankings

Top 10 Counties

Rank	County	Predicted Livability Score	Category
1	Chester	86.49	Excellent
2	Montgomery	85.09	Excellent
3	Bucks	80.45	Excellent
4	Cumberland	76.67	Excellent
5	Butler	74.72	Good
6	Delaware	72.21	Good
7	Westmoreland	70.13	Good
8	Allegheny	69.39	Good
9	Northampton	67.52	Good
10	Washington	67.17	Good

All Counties

Rank	County	Predicted Livability Score	Category
67	Potter	38.6	Needs Improvement
66	Philadelphia	36.28	Needs Improvement
65	Fayette	38.18	Needs Improvement
64	Cameron	38.26	Needs Improvement
63	Millin	40.08	Needs Improvement
62	Clinton	44.31	Needs Improvement
61	Jefferson	45.11	Average
60	Juniata	45.53	Average
59	Sullivan	45.62	Average
58	Potter	45.72	Average

Download County Rankings as CSV

Model Details:

- This tab shows the selected model, prediction target, number of features, model comparison results, feature list, and evaluation output.

County Selection

Select a Pennsylvania County

Philadelphia

Project Theme: Best in PA

ML Task: Regression

Model: Machine Learning Model

Output: Livability Score

Deploy

Best PA County to Live In Predictor

This web app uses a trained machine learning model to predict and rank Pennsylvania counties based on livability-related indicators. The score is based on economic, health, education, housing, and community access factors.

County PredictionRankingsModel DetailsProject Notes

Model Details

Model Type: Machine Learning Model

Prediction Target: Livability Score

Number of Counties: 67

Number of Model Features: 21

Model Comparison

Model	MAE	RMSE	R ² Score
Linear Regression		0	0
Random Forest Regressor	2.9894		3.6315
Decision Tree Regressor	4.3888		5.1822

Features Used by the Model

Feature Column
unemployment_rate
children_poverty
severe_housing_problems
uninsured_rate
poor_or_fair_health
poor_physical_health_days
poor_mental_health_days
income_inequality
child_care_cost_burden
rural_percent

Model Evaluation Output

Model Training Summary

```
.....
```

Project Notes:

- This tab explains the project idea, the data sources, and how the livability score was created.

County Selection

Select a Pennsylvania County

Philadelphia

Project Theme: Best in PA

ML Task: Regression

Model: Machine Learning Model

Output: Livability Score

Deploy

Best PA County to Live In Predictor

This web app uses a trained machine learning model to predict and rank Pennsylvania counties based on livability-related indicators. The score is based on economic, health, education, housing, and community access factors.

County PredictionRankingsModel DetailsProject Notes

Project Notes

This project follows the "Best in PA" theme by defining the best county as a county with strong livability indicators. The model uses public Pennsylvania county-level data and creates a livability score through feature engineering.

The project combines two public data sources: the 2025 Pennsylvania County Health Rankings workbook and the PA Open Data ACS income and Poverty county dataset. These sources provide information about income, poverty, health, education, housing, broadband access, population, and community conditions.

The livability score combines five main areas: economic strength, health, education, community access, and housing/family conditions. Multiple regression models were compared, and the final app uses the best-performing model based on the evaluation results.

The selected final model is **Machine Learning Model**. This application is designed for inference. Users can select a county and immediately view the predicted score, state rank, category, and supporting county indicators.

7. Discussion and Conclusion

The results show that counties such as Chester, Montgomery, Bucks, and Cumberland ranked near the top based on our livability score. These counties generally had strong results in areas like income, education, health, and community access.

One important thing we learned is that the ranking depends on how “livability” is defined. Since we created our own score, the result is based on the features and weights we chose. If we changed the weights or added other features, such as crime rate, school ratings, housing cost, or environmental risk, the ranking could change.

One limitation of the project is that Pennsylvania only has 67 counties, so the final model does not have a very large number of county records. Another limitation is that the target score was engineered by us instead of coming from an official livability ranking. However, this also fits the project theme because the assignment allows students to define what “best” means in Pennsylvania.

Overall, this project used many course concepts, including data collection, data cleaning, feature engineering, regression models, model evaluation, and web app development. The final app successfully connects a trained machine learning model to a simple user interface, which makes the project more practical than only showing the model in a notebook.

In conclusion, we were able to build a working machine learning web application that predicts and ranks Pennsylvania counties based on livability indicators. The app gives users a simple way to explore county rankings and understand the factors behind the prediction.

8. AI Usage

AI tools were used as a support tool during this project. ChatGPT helped with brainstorming the project idea, organizing the folder structure, debugging Python virtual environment problems, improving the Streamlit app, helping organize the final report, and checking grammar.